

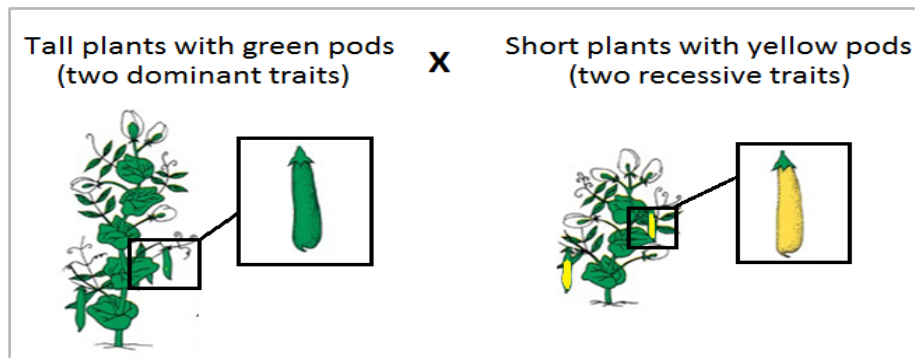
Name: _____

MENDELIAN GENETICS

Topic 2: Dihybrid Crosses

Mendel also conducted experiments to determine whether the pattern of segregation of alleles for one gene had any influence on that of another gene.

For example, is the height of a pea plant related to the colour of pods it produces? In order to determine whether these two traits are linked, two true-breeding plants for two different traits were crossed...



Parental phenotype		
Parental genotype		
Gametes produced		

This cross yielded the following results:

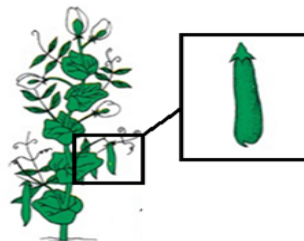
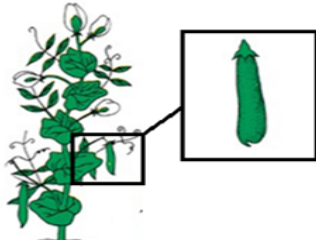
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Like in the F₂ generation of Mendel's monohybrid crosses, the crossing of F₁ individuals produced a predictable phenotypic ratio in the F₂ generation of dihybrid crosses as well...

Tall plants with green pods
(two dominant traits)

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Tall plants with green pods
(two dominant traits)



F1 phenotype		
F1 genotype		
Gametes produced		

This cross yielded the following results:

Because such crosses consistently yielded this 9:3:3:1 ratio, Mendel concluded that the segregation of alleles for one gene had no influence on the segregation of alleles for another gene. This is known as the Law of Independent Assortment.

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PRACTICE: DIHYBRID CROSSES

1) In tomatoes, red fruit (R) is dominant to yellow fruit (r), and tall plants (T) are dominant to short plants (t). A plant that is heterozygous for both traits is crossed with a plant that is heterozygous for the gene encoding height, yet produces yellow fruit.

a. What are the phenotypes of the plants being crossed?

b. What are the genotypes of the plants being crossed?

c. Construct a Punnett Square to represent this cross.

d. What proportion of offspring will be...

- Tall with red fruit?
- Tall with yellow fruit?
- Short with red fruit?
- Short with yellow fruit?